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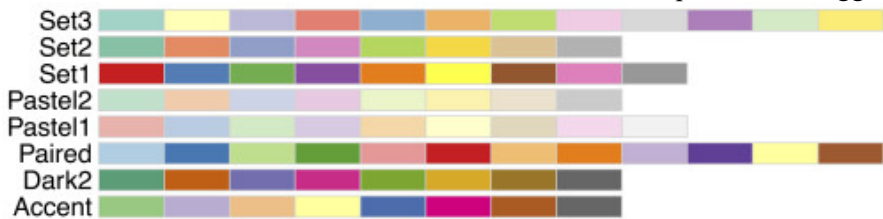
Create Colorful Graphs in R with RColorBrewer and Plotly

Published ~~April 14, 2015~~[January 4, 2016](#) by [chelsea](#) in [R](#)

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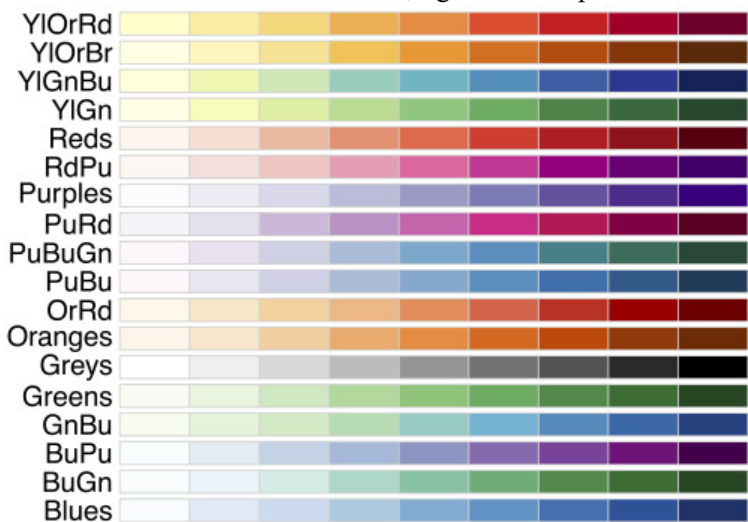
[RColorBrewer](#) is an R package that allows users to create colourful graphs with pre-made color palettes that visualize data in a clear and distinguishable manner. There are 3 categories of palettes: qualitative, diverging, and sequential.

- Qualitative palettes** employ different hues to create visual differences between classes. These palettes are suggested



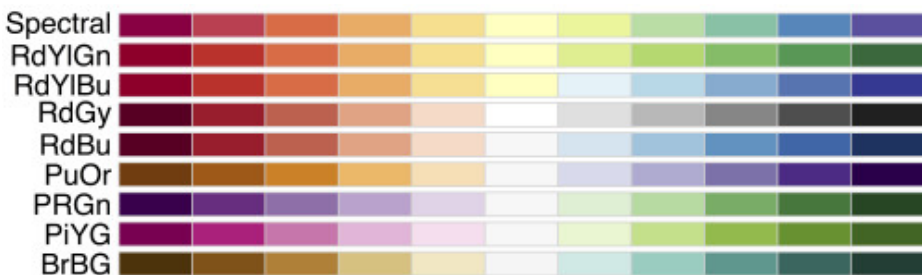
for nominal or categorical data sets.

- Sequential palettes** progress from light to dark. When used with interval data, light colors represent low data values



and dark colors represent high data values.

- Diverging palettes** are composed of darker colors of contrasting hues on the high and low extremes and lighter colors in the middle.



The palettes are composed of 8-12 distinct colors, but if you have more than 12 categories to graph, you can use the `colorRampPalette()` function with any of the sequential or diverging palettes. This ramps the color at the necessary

interval to create as many hues as your data calls for.

Below are a few examples using RColorBrewer and Plotly! The code for the first example is provided below and the full R scripts and data for the second and third examples are available [here](#).

1) Bar Graph with the Paired palette:

First, add the libraries:

```
1 library(RColorBrewer)
2 library(plotly)
3 py <- plotly()
```

Add x & y data and define the type of graph as "bar". To use RColorBrewer define

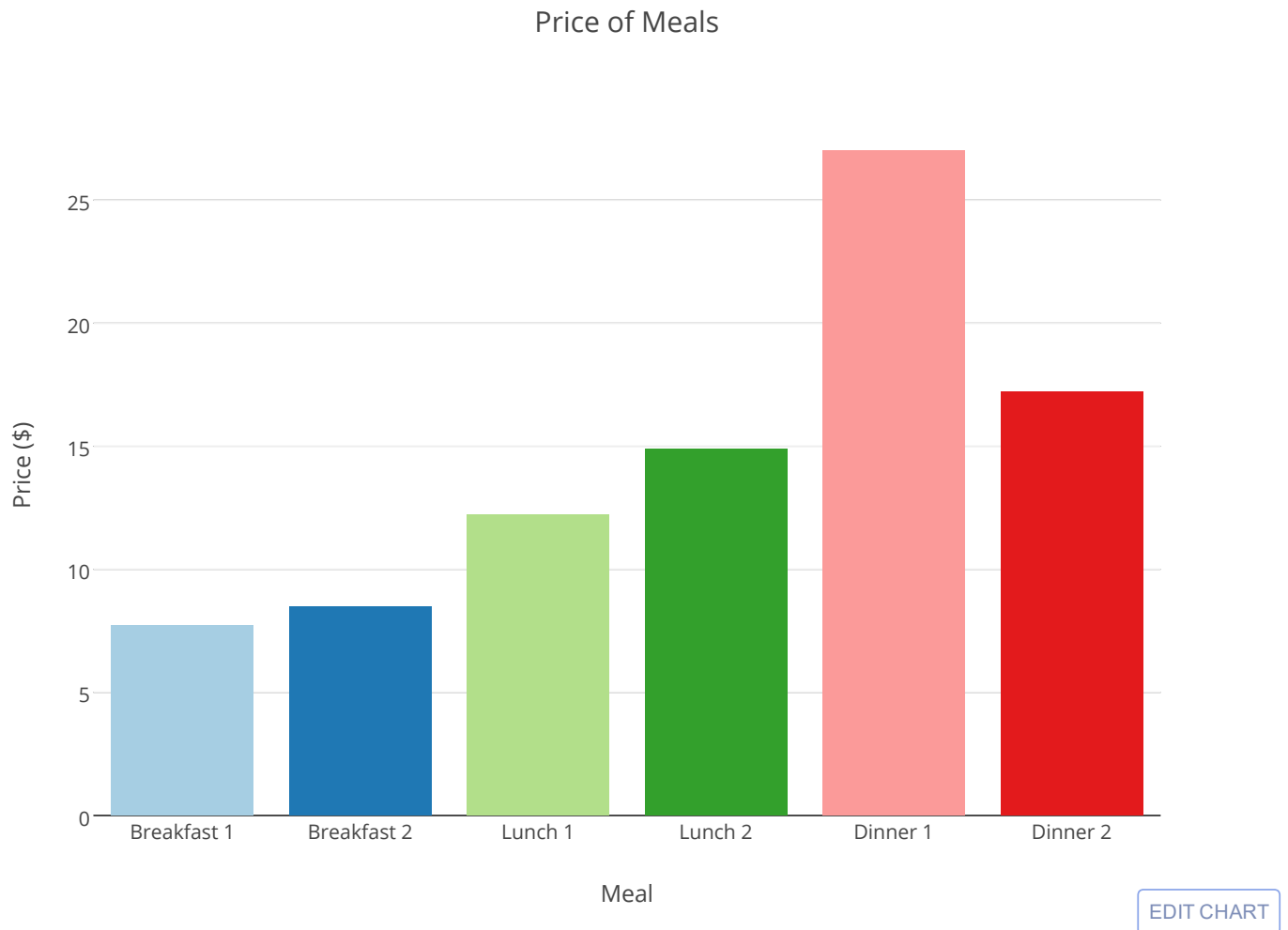
```
1 marker = list(color = brewer.pal(n_palette, "Palette_Name"))
```

where n_palette is the number of colors from the palette that you want to use in the graph.

```
1 data_bar <- list(x = c("Breakfast 1", "Breakfast 2", "Lunch 1", "Lunch 2", "Dinner 1", "Dinner 2"),
2                  y = c(7.72, 8.5, 12.22, 14.89, 27.02, 17.23),
3                  type = "bar",
4                  marker = list(color = brewer.pal(6, "Paired")))
5 )
```

Finally, specify any layout information and send to Plotly!

```
1 layout_bar <- list(title = "Price of Meals", xaxis = list(title = "Meal"), yaxis = list(title = "Price"),
2 response <- py$plotly(data = data_bar, kwargs = list(layout = layout_bar, filename = "Your_File"))
```



2) Scatterplot with `colorRampPalette()` and the Spectral palette:

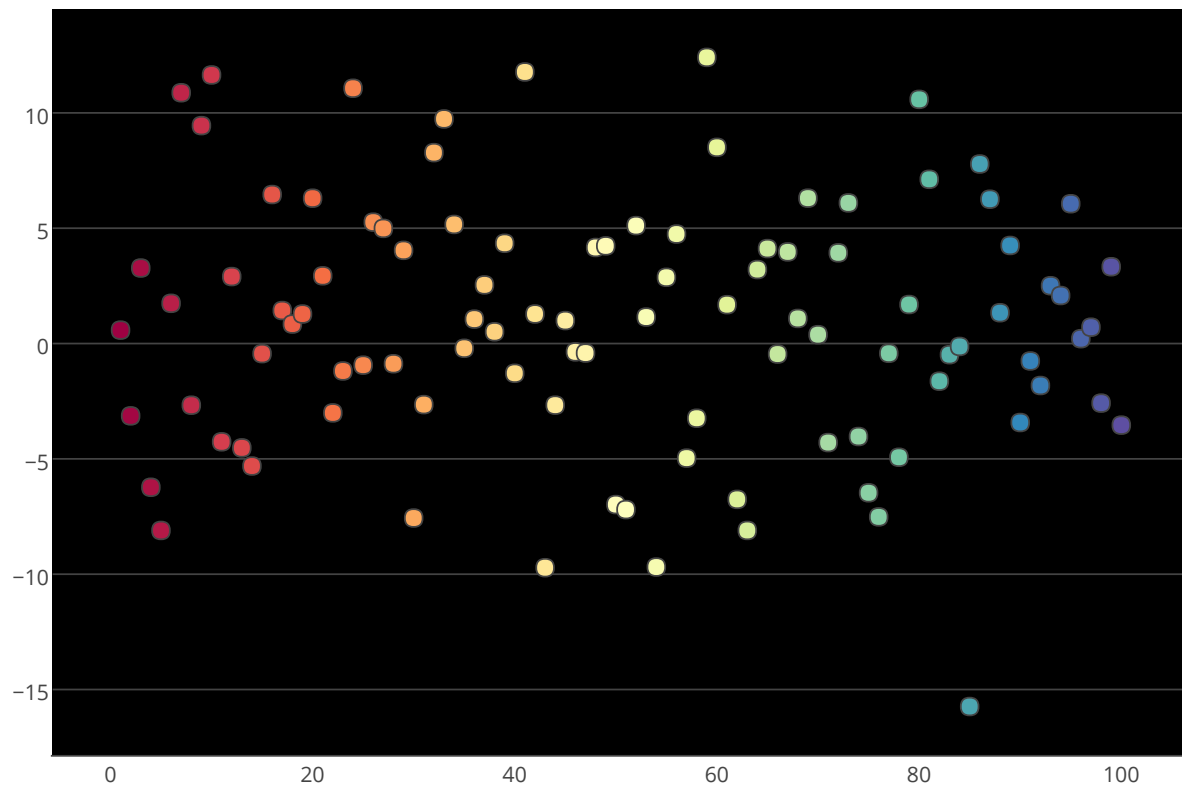
The key here is to set

```
colorRampPalette(brewer.pal(n_palette, "palette_name"))(n_plot),
```

where `n_palette` is the number of colors from the palette that you want to use and `n_plot` is the number of colors you want in your plot. For this example, we'll use the Spectral palette. The Spectral palette has a max of 11 colors. We'll use the full palette and we want each of the 100 points to be a different color so:

```
1 marker = list(color = colorRampPalette(brewer.pal(11, "Spectral"))(100))
```

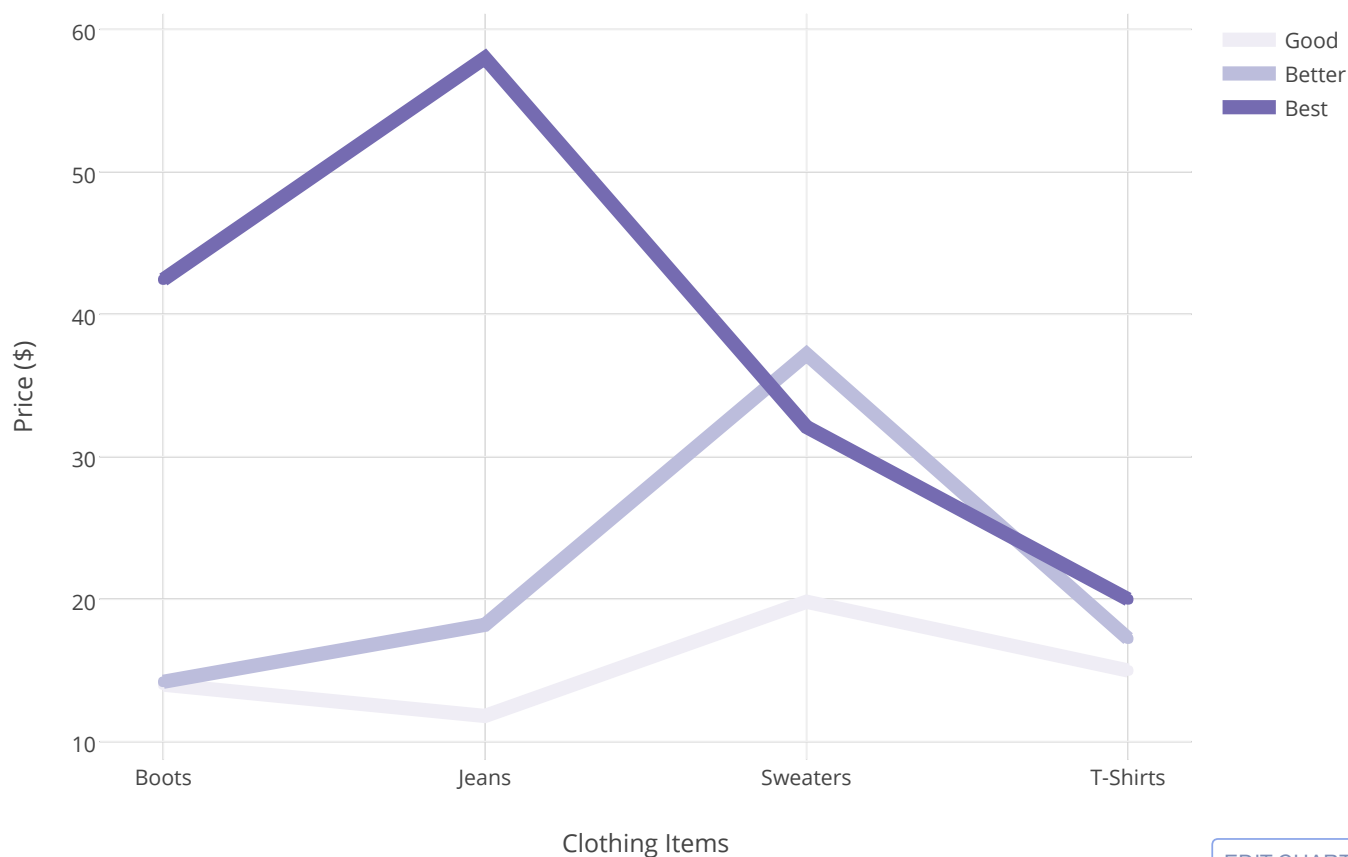
ROYGB Scatter

[EDIT CHART](#)**3) Line Graph with ggplot syntax:**

You can also use RColorBrewer with ggplot with the command `scale_colour_brewer(palette = "Palette_Name")`. You can view the full [script](#) for this example, but the generic syntax is:

```
1 ggplot(data = Your_Data,  
2       aes(x = X_Variable,  
3         y = Y_Variable,  
4         group = Group_Variable,  
5         colour = Group_Variable)) +  
6 scale_colour_brewer(palette = "Palette_Name")  
7  
8 py$ggplotly(kwarg = list(filename = "Your_Filename",  
9                          fileopt = "overwrite")  
10 )
```

Price and Quality of Clothing Items

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**chelsea**Tags: [color](#), [ggplot](#), [Plotly](#), [R](#), [RColorBrewer](#)

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